

Experiment Trials

~Jerome Grant, Matu Saye, Samuel Duro-Aina

Trial A: Map-Reduce Test

Settings

- TSshark ready
- Model Selection
 - Manager: Rama3.8b
 - Researcher: deepseekv1.7b
 - Editor: qwen2.5.7b
- Network
 - Interface: Wi-Fi
- Packets: 15
- Session ID: session_202620
- Live Capture
 - CAPTURE NOW
- Last:

Identify the packets from the live traffic capture for IP addresses that may be doing suspicious activities, then give me a status report on them.

researcher

Threat Hunter LIVE TRAFFIC CAPTURE:

INTERFACE: Wi-Fi PACKETS: 15

- 192.168.1.207 → 142.251.151.119 Protocol: UDP
- 142.251.151.119 → 192.168.1.207 Protocol: UDP
- 7 → 7 Protocol: ARP
- 7 → 7 Protocol: ARP
- 192.168.1.207 → 172.253.122.188 Protocol: TCP
- 172.253.122.188 → 192.168.1.207 Protocol: TCP
- 192.168.1.207 → 142.251.151.119 Protocol: UDP
- 192.168.1.207 → 142.251.151.119 Protocol: UDP
- 192.1
- 3.85s

editor

Analyst Feedback: The request is clear but could benefit from some additional details to ensure accuracy and completeness. For example:

- What specific types of suspicious activities are you looking for (e.g. 27.2s)

manager

Commander Suspicious Activity Report

After analyzing the live traffic capture, I have identified several packets that may be indicative of suspicious activities:

- Packet 1-2:** The exchange between IP addresses 192.168.1.207 and 142.251.151.119 appears to be a potential command and control (C2) communication, as it uses UDP protocol. This type of communication is often used by malware or attackers to send commands or receive data.
- Packet 5-6:** The TCP connection between IP addresses 192.168.1.207 and 172.253.122.188 appears to be a potential data transfer, as it uses TCP protocol. This type of communication is often used by malware or attackers to transfer data.

117.75s

Ask about cybersecurity (e.g., "Show live traffic", "Check IP 185.130.5.253")

Latency:

Researcher-3.65s

Editor-27.2s

Manager-117.75

In the trial I asked the assistant to identify packets from the live capture results with IP addresses that may be doing suspicious activity and to give a status report on them. Upon observation, although running in parallel, the researcher and the editor nodes did not finish at the same time. The researcher node finished the fastest with 3.65s latency while the editor completed its task with a latency of 27.2s. What the reducer does is take the outputs, which are the IP packets and the editor deals with the context of the user query, in order to combine them into a final answer or report.

Trial B: Human-in-the-Loop Test

The screenshot displays a Human-in-the-Loop (HIL) test interface. On the left is a control panel for a network analysis tool, and on the right is a chat interface for human interaction.

Control Panel (Left):

- Model Selection:** llama3.8b (dropdown), Researcher: deepseek-r1.7b (dropdown), Editor: qwen2.5:7b (dropdown).
- Network Interface:** Wi-Fi (dropdown).
- Packets:** A slider set to 10.
- Session ID:** session_232424.
- Live Capture:** Interface: Wi-Fi, CAPTURE NOW button.
- Last:** INTERFACE: Wi-Fi, PACKETS: 10. A list of captured packets is shown below.

Chat Interface (Right):

- Initial Prompt:** "Explain the pros and cons of using a Null Scan on a windows machine".
- Response:** A detailed response explaining the pros and cons of Null Scan, including early detection of vulnerabilities and maintenance of system integrity.
- User Feedback:** "Packets 5-10 were not mentioned. Rewrite the detailed summary to include all the packets from the live capture results".
- Revised Response:** A revised summary of the live capture results, including details about the captured data, flow 1, and flow 2.

Packet List (from 'Last' section):

No.	Source IP	Destination IP	Protocol
1.	142.250.188.10	192.168.1.1	UDP
2.	192.168.1.207	142.250.188.1	UDP
3.	?	?	?

The user provided human-in the loop feedback stating packets 5-10 were not mentioned and asked the manager node to reroute the task back to the researcher(threat hunter) and the editor(analyst) node. After receiving revised data from Editor and researcher nodes, it generated a comprehensive response which produced 1-8 packets due to it's output constraints. Below is the screenshot.

 Packets 5-10 are not mentioned. Rewrite the detailed summary to include all the packets from the live capture results

 researcher

 Threat Hunter  LIVE TRAFFIC CAPTURE:

INTERFACE: Wi-Fi PACKETS: 10

1. 192.168.1.207 → 142.250.188.10 Protocol: UDP
2. 142.250.188.10 → 192.168.1.207 Protocol: UDP
3. 13.69.239.74 → 192.168.1.207 Protocol: TCP
4. 192.168.1.207 → 13.69.239.74 Protocol: TCP
5. 192.168.1.207 → 142.251.167.189 Protocol: UDP
6. 142.251.167.189 → 192.168.1.207 Protocol: UDP
7. 192.168.1.207 → 162.159.130.233 Protocol: SSL
8. 192.168.1.207 →

 3.28s

 editor

 Analyst APPROVED

If you need any specific guidance on how to rewrite the summary, please let me know!

 8.76s

 manager

 **Commander** Here is the rewritten detailed summary to include all the packets from the live capture results:
Analysis:  LIVE TRAFFIC CAPTURE:

INTERFACE: Wi-Fi PACKETS: 10

1. 192.168.1.207 → 142.250.188.10 Protocol: UDP
2. 142.250.188.10 → 192.168.1.207 Protocol: UDP
3. 13.69.239.74 → 192.168.1.207 Protocol: TCP
4. 192.168.1.207 → 13.69.239.74 Protocol: TCP
5. 192.168.1.207 → 142.251.167.189 Protocol: UDP
6. 142.251.167.189 →

 122.03s

 interrupt

Trial C: Quantization Test

Q8 response from Assistant

The screenshot displays the 'Cybersecurity Incident Response Agent' interface. On the left is a sidebar with sections: 'Model Selection' (Manager: llama3.8b-instruct-q8_0, Researcher: deepseek-r1.7b, Editor: qwen2.5.7b), 'Network' (Interface: Wi-Fi, Packets: 10), 'Session ID' (session_004322), 'Live Capture' (Interface: Wi-Fi, CAPTURE NOW button), and 'Metrics' (Editor: 28.48s, Researcher: 191.1s, Manager: 130.52s). The main area shows a chat log with the following messages:

- editor** (28.48s): Analyst Feedback: Please provide the original response you would like me to verify, so I can ensure it's accurate before giving my approval.
- researcher** (191.1s): Threat Hunter Null Scan on Windows Machines: Pros and Cons
Pros:
 1. **No Prior Knowledge Required:** Null Scans are straightforward as they don't require the attacker to know specific port numbers or vulnerabilities beforehand, making them quick to execute.
 2. **Efficient Execution:** The process stops once a responsive port is found, minimizing unnecessary attempts and reducing time spent on ineffective scans.
 3. **Stealthy Nature:** Since it can be automated without user interaction, it poses few
- manager** (130.52s): **Commander Final Response:**
A Null Scan on a Windows machine can be a useful tool for attackers, but it also has its drawbacks. Here are the key pros and cons to consider:
Pros:
 1. **No Prior Knowledge Required:** Null Scans are straightforward as they don't require the attacker to know specific port numbers or vulnerabilities beforehand, making them quick to execute.
 2. **Efficient Execution:** The process stops once a responsive port is found, minimizing unnecessary attempts and reducing time s
- Interrupt**

Latency:

Editor -28.48s

Researcher-191.1s

Manager-130.52s

Note: final response does not complete.

Final Response:

The screenshot displays the 'Cybersecurity Incident Response Agent' interface. On the left is a sidebar with sections: 'Settings' (TShark ready), 'Model Selection' (Manager: llama3.8b-instruct-q8_0, Researcher: deepseek-r1.7b, Editor: qwen2.5.7b), and 'Metrics'. The main area shows a chat log with the following messages:

- Final Response:**
A Null Scan on a Windows machine can be a useful tool for attackers, but it also has its drawbacks. Here are the key pros and cons to consider:
Pros:
 1. **No Prior Knowledge Required:** Null Scans are straightforward as they don't require the attacker to know specific port numbers or vulnerabilities beforehand, making them quick to execute.
 2. **Efficient Execution:** The process stops once a responsive port is found, minimizing unnecessary attempts and reducing time spent on ineffective scans.
Cons:
 1. **Inaccurate Results:** Null Scans can produce false posit

Versus

Q2 response from Assistant

The screenshot displays the 'Cybersecurity Incident Response Agent' interface. On the left, a sidebar contains controls for Model Selection (Manager: llama3.8b-instruct-q2_K, Researcher: deepseek-r1.7b, Editor: qwen2.5.7b), Network (Interface: Wi-Fi, Packets: 10), Session ID (session_225255), Live Capture (CAPTURE NOW), and Metrics (Editor: 28.34s, Researcher: 164.58s, Manager: 65.96s). The main panel shows a chat history with a prompt 'Explain the pros and cons of using a Null Scan on a windows machine'. The chat includes an 'Analyst Feedback' message, a 'Threat Hunter' message, and a 'Commander' message. The 'Commander' message provides a final response titled 'The Pros and Cons of Using a Null Scan on a Windows Machine', listing benefits like efficiency, smooth user experience, and early threat detection. The interface also features a 'System Info' link and a search bar at the bottom.

Latency:

Editor-28.34s

Researcher-164.58s

Manager-65.96s

Note: final response does not complete.

Final Response:

This screenshot shows the same interface as the previous one, but with the final response displayed. The 'Commander' message now includes a section titled 'However, there are also some drawbacks to consider:' followed by a list of drawbacks. The interface elements, including the sidebar controls and chat history, remain the same.

Upon observation, the 2-bit model of llama 3:8b loses the logical 'thread' when compared to the 8-bit version of llama3:8b, especially when it comes to completing the task and maintaining coherence. The 2-bit version of the model hallucinated the pros of a Null Scan as well as failed to completely provide a con for a Null Scan. In the 8-bit version of the model, the final response was coherent as a Null Scan is indeed a useful tool for hackers and was able to provide 2 pros and almost completed a con. On the brighter side, at least it was able to maintain coherence.